

Introduction to Knot Theory

(3) Alexander Polynomial

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Goal: A Polynomial Invariant of Knots

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Examples

Knot	Polynomial
Unknot	1
Trefoil	$t^{-1} - 1 + t$
Figure-eight	$-t^{-1} + 3 - t$

Goal: A Polynomial Invariant of Knots

Today's question

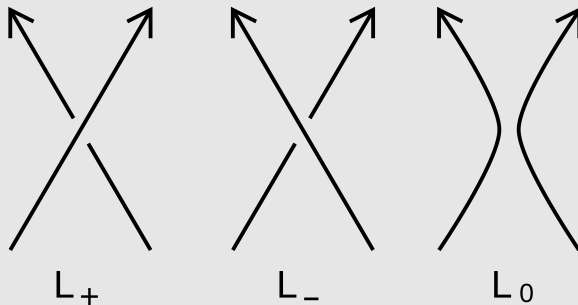
How do we **compute** this polynomial for any given knot?

Conway's Skein Relation: Setup

Key observation: Changing just one crossing gives a different knot.

Conway's idea: *relate* the polynomial of a knot to simpler ones obtained this way.

Three diagrams, identical everywhere except at one crossing



Conway Polynomial: Definition

Definition (Conway, 1970)

The **Conway polynomial** $\nabla_K(z) \in \mathbb{Z}[z]$ is uniquely determined by:

C1 $\nabla(\bigcirc) = 1$

C2 $\nabla(L_+) - \nabla(L_-) = z \nabla(L_0)$

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How to use it:

- Pick any crossing in the diagram
- Write the skein relation to express ∇ in terms of simpler diagrams
- Repeat until everything reduces to the unknot
- No matrix, no group theory needed

Note

$\nabla_K(z)$ is a polynomial in z with integer coefficients.

All the knot information is encoded in a single variable z .

Example: Conway Polynomial of the Trefoil

Goal: Compute $\nabla(\text{trefoil})$ step by step.

At one crossing: $L_+ = \text{trefoil}$, $L_- = \text{unknot}$, $L_0 = \text{a simpler diagram}$.

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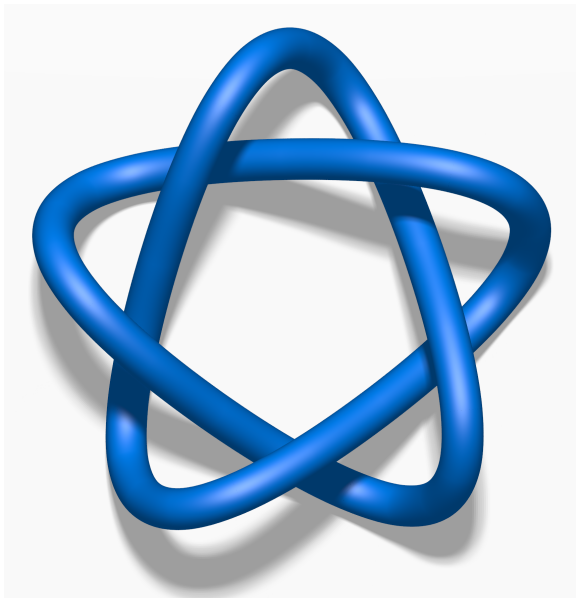
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$\nabla(\text{trefoil}) = 1 + z^2 \neq 1 = \nabla(\text{unknot})$, so the trefoil is not the unknot. ✓

Example: Conway Polynomial of Cinquefoil knot(5_1 knot)



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Theorem

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The **Alexander polynomial** $\Delta_K(t) \in \mathbb{Z}[t, t^{-1}]$ is obtained from $\nabla_K(z)$ by the substitution $z = t^{1/2} - t^{-1/2}$.

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Check on the trefoil:

$$\nabla(\text{trefoil}) = 1 + z^2 \xrightarrow{z = t^{1/2} - t^{-1/2}} 1 + (t^{1/2} - t^{-1/2})^2 = 1 + (t - 2 + t^{-1}) = t^{-1} - 1 + t$$

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Conway $\nabla_K(z) \in \mathbb{Z}[z]$

- Easy to compute recursively
- No matrix needed

Alexander $\Delta_K(t) \in \mathbb{Z}[t, t^{-1}]$

- Direct topological meaning
- Defined via a matrix (see next)

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Apply **Fox calculus** — a formal partial derivative on the group ring — to each relator:

$$A_{ij} = \left. \frac{\partial r_i}{\partial x_j} \right|_{x_k \mapsto t} \in \mathbb{Z}[t, t^{-1}]$$

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Definition (Alexander Polynomial)

$$\Delta_K(t) := \det A'$$

where A' is A with any one column deleted.

The result is well-defined up to units $\pm t^k \in \mathbb{Z}[t, t^{-1}]$.

Fox Calculus: Rules

Fox Derivative $\partial/\partial x_j : \mathbb{Z}[F_n] \rightarrow \mathbb{Z}[F_n]$

- $\frac{\partial x_i}{\partial x_j} = \delta_{ij}$
- **Leibniz rule:** $\frac{\partial(uv)}{\partial x_j} = \frac{\partial u}{\partial x_j} + u \frac{\partial v}{\partial x_j}$
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After differentiating each relator r_i with respect to each generator x_j ,
abelianize by sending every $x_k \mapsto t$.
The entries of the resulting matrix lie in $\mathbb{Z}[t, t^{-1}]$.

Example: Trefoil via the Alexander Matrix

Wirtinger presentation: $G(\text{trefoil}) = \langle x_0, x_1, x_2 \mid r_i = x_{i+1}^{-1} x_{i-1} x_{i+1} x_i^{-1} \rangle$ (indices mod 3)

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Same answer as Conway's method. ✓

(They must agree by the theorem $\Delta_K(t) = \nabla_K(t^{1/2} - t^{-1/2})$.)

Example: Trefoil via Simplified Presentation

Alternative approach: Use a simplified knot group presentation.

$$G(\text{trefoil}) = \langle x, y \mid xyx = yxy \rangle \quad \Rightarrow \quad r = xyxy^{-1}x^{-1}y^{-1}$$

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$$\frac{\partial r}{\partial x} = 1 + xy - xyxy^{-1}x^{-1}$$

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Result

Delete one column (e.g., col 2):

$$\Delta_K(t) \doteq \det(1 - t + t^2) = \mathbf{t^2 - t + 1}$$

Limitations of the Alexander Polynomial

The Alexander polynomial is *not* a complete invariant

Two distinct knots can share the **same** Alexander polynomial.

- There exist non-trivial knots with $\Delta_K(t) = 1$
- Conway knot, Kinoshita–Terasaka knot

$$\Delta_K(t) = 1, \quad K \neq \bigcirc$$

- So $\Delta_K(t) = 1$ does *not* imply K is the unknot

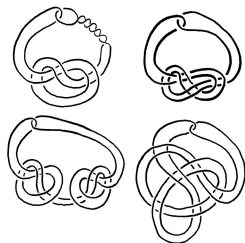


Figure: A knot with $\Delta_K(t) = 1$ that is not the unknot

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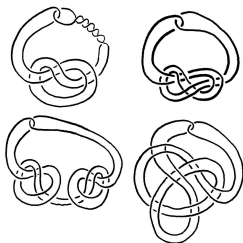


Figure: A knot with $\Delta_K(t) = 1$ that is not the unknot

Stronger invariants are needed $\rightarrow$ Jones polynomial, HOMFLYPT polynomial, ...

Summary

Two Ways to Compute $\Delta_K(t)$

① **Conway skein relation (easy)**

$$\nabla(L_+) - \nabla(L_-) = z \nabla(L_0)$$

$$\text{then } \Delta_K(t) = \nabla_K(t^{1/2} - t^{-1/2})$$

② **Alexander matrix (original)**

Wirtinger $\xrightarrow{\text{Fox calculus}}$ matrix A

$$\Delta_K(t) = \det A'$$

Key Properties

- $\Delta_K(1) = 1$ (normalization)
- $\Delta_K(t) = \Delta_K(t^{-1})$ (symmetry)
- **Not** a complete invariant

The big picture

Conway found a **recursive shortcut** for the same invariant Alexander defined via matrices in 1928.

References

- J. W. Alexander, “Topological invariants of knots and links,” *Trans. Amer. Math. Soc.* 30 (1928), 275–306.
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Images

- Rolfsen, *Knots and Links*